

Detecting Silent Failures in Imitation Learning Rollouts with Contrastive TAP-Score

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Abstract

Imitation learning policies can fail silently under distribution shift: the policy continues producing actions even when the state drifts outside the training manifold. This report studies failure detection for closed-loop visuomotor imitation policies. We identify Diffusion Policy as a strong baseline for Push-T and replicate rollouts using a downloaded checkpoint. We then propose and implement TAP-Score, a failure monitor trained only on expert demonstrations using a contrastive ranking objective (InfoNCE) to avoid binary classifier collapse. We integrate TAP-Score into Diffusion Policy rollouts and evaluate detection on clean and observation-perturbed executions with paired seeds. Matching TAP-Score’s action chunk horizon to the policy horizon (8-step chunks) improves success-versus-failure detection from AUROC 0.553 (stitched baseline) to 0.690 (matched), while preserving strong clean-versus-perturbed separation (AUROC about 0.8).

1 Problem

In imitation learning, a policy $\pi(a_t \mid o_{\leq t})$ is trained from expert demonstrations to map observations to actions. Under distribution shift (lighting changes, occlusions, unusual object placement), policies may fail while remaining confident. The goal is to detect failure early, using only expert data for training the detector.

We seek a scoring function $S(o_{t:t+T-1}, a_{t:t+H-1})$ that is high for expert-like proposals and low for off-manifold proposals, enabling an intervention rule:

if $S < \tau$ then intervene.

2 SOTA: Diffusion Policy

We identify Diffusion Policy [2] as a strong baseline for visuomotor imitation learning. In Push-T, the policy conditions on a short observation history window ($T = 2$ frames) and predicts an action chunk of length $H_{\text{DP}} = 8$. In closed loop, it replans each policy step and executes $n_{\text{action}} = 8$ actions per plan (receding-horizon execution).

3 Replication: Running a Downloaded Checkpoint

We replicate evaluation by loading a downloaded Diffusion Policy checkpoint and running paired-seed rollouts in Push-T. We run 50 seeds (0 to 49) for clean, noise, and blur conditions, using identical seeds across conditions for fair comparison.

Table 1: Diffusion Policy rollout performance on Push-T (50 seeds, paired across conditions).

Condition	Episodes	Success Rate	Mean Reward
Clean	50	0.54	0.677
Noise	50	0.48	0.616
Blur	50	0.54	0.671

4 Proposed Idea: Contrastive TAP-Score

4.1 Why Ranking Instead of Binary Classification

A naive monitor trains a binary classifier to label expert pairs (o, a) as positive and random pairs as negative. In practice, binary training can collapse toward observation-agnostic predictions. We instead frame the task as ranking: given one expert action and M negatives, predict which action is expert.

4.2 Model and Objective

Let $f(o) \in \mathbb{R}^d$ be an observation encoder and $g(a) \in \mathbb{R}^d$ an action encoder, both ℓ_2 normalized. Similarity is cosine:

$$\ell(o, a) = \frac{f(o)}{\|f(o)\|_2} \cdot \frac{g(a)}{\|g(a)\|_2}.$$

Given a positive chunk a^+ and negatives $\{a_i^-\}_{i=1}^M$, InfoNCE [3, 1] is

$$\mathcal{L} = -\log \frac{\exp(\ell(o, a^+)/\tau)}{\exp(\ell(o, a^+)/\tau) + \sum_{i=1}^M \exp(\ell(o, a_i^-)/\tau)}.$$

4.3 Inference Score

We score a proposed action chunk a relative to negatives:

$$S(o, a) = \ell(o, a) - \log \sum_{i=1}^M \exp(\ell(o, a_i^-)).$$

Lower scores indicate the proposal is less distinguishable from random alternatives.

5 Implementation

5.1 Integration with DP Rollouts

For each policy step in a rollout we log:

- observation window $o_{t:t+1}$ (two frames),
- the policy-proposed 8-step action chunk,
- rollout rewards and the final success label (max reward threshold).

We then run TAP-Score on each logged step and aggregate per-episode scores.

5.2 Two TAP Variants

We evaluate two monitors:

- H=16 stitched: TAP expects 16 actions, but DP proposes 8, so we stitch two consecutive 8-step proposals.
- H=8 matched: TAP expects 8 actions, matching DP proposals directly (no stitching).

5.3 Episode Aggregations

We use two episode-level scores:

- p10: 10th percentile of per-timestep TAP scores (smooth ranking).
- min: minimum per-timestep TAP score (trigger-style monitoring).

5.4 Calibration and Operating Points

We calibrate the intervention threshold τ on **clean success episodes only**. For a target false positive rate (FPR) of 1%, 5%, 10%, we select τ as the corresponding percentile of the calibration distribution of episode min-scores.

We report:

- FPR(calib): measured FPR on clean successes (the calibration set).
- FPR(all successes): measured FPR on all successful episodes (including successful perturbed episodes).

6 Results

6.1 Clean vs Perturbed (distribution shift detection)

TAP separates clean from perturbed rollouts with AUROC around 0.8 in both baselines. This indicates the monitor is sensitive to observation distribution shift.

6.2 Success vs Failure (failure detection)

H=8 matched improves success-versus-failure separation compared to H=16 stitched.

6.3 Operating Point Details for Failure Detection

At the 5% target operating point, the trigger rule flags an episode if its minimum per-timestep score falls below τ .

6.4 Early Warning Behavior (H=8 matched)

Among flagged failure episodes ($n = 53$), the mean detection index is 6.8 policy steps and mean remaining steps after trigger is 17.2 policy steps. Since each policy step executes 8 environment actions, this corresponds to roughly $17.2 \times 8 \approx 138$ environment actions of lead time. 88.7% of flagged failures are triggered before 70% of the episode horizon.

Table 2: Detection summary on Push-T (50 clean, 50 noise, 50 blur; paired seeds; clean-only calibration; trigger-style evaluation).

Metric	H=16 stitched	H=8 matched
Clean vs perturbed AUROC (p10)	0.773	0.812
Clean vs perturbed AUROC (min)	0.804	0.795
Success vs failure AUROC (p10)	0.553	0.690
Success vs failure AUROC (min)	0.607	0.661
TPR at 5% target (success/failure)	63.9%	73.6%
Score separation (success mean p10 vs failure mean p10)	-4.09 vs -4.45	-2.48 vs -3.36
Failures flagged early (before 70%)	93.5%	88.7%

Table 3: 5% target operating point (trigger rule using episode min-score), success vs failure detection.

Variant	τ	TPR	FPR(calib)	FPR(all successes)
H=16 stitched	-4.175	63.9%	7.4%	42.3%
H=8 matched	-3.180	73.6%	7.4%	34.6%

7 Discussion

Key takeaway. Matching the TAP action chunk horizon to the policy horizon materially improves failure detection metrics. In particular, success-versus-failure AUROC (p10) improves from 0.553 to 0.690, and trigger-style TPR at the 5% target improves from 63.9% to 73.6%.

Calibration limitation. While thresholds are calibrated on clean successes, the measured false alarm rate on all successful episodes is higher, indicating that a deployment-grade monitor would benefit from refined calibration (for example, per-condition calibration or a two-stage trigger).

DP checkpoint limitation. The evaluated DP checkpoint achieves 54% clean success on this setup, leaving room for a stronger baseline that may sharpen success versus failure separation.

8 Conclusion

We implemented and evaluated TAP-Score, a contrastive action-proposal monitor trained only on expert demonstrations, integrated with closed-loop Diffusion Policy rollouts on Push-T. Using paired seeds and clean-success calibration, we find that horizon-matched TAP (H=8) improves success-versus-failure detection and provides early warning signals with substantial lead time.

References

- [1] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. *arXiv preprint arXiv:2002.05709*, 2020.
- [2] Cheng Chi et al. Diffusion policy: Visuomotor policy learning via action diffusion. *arXiv preprint arXiv:2303.04137*, 2023.

- [3] Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748*, 2018.